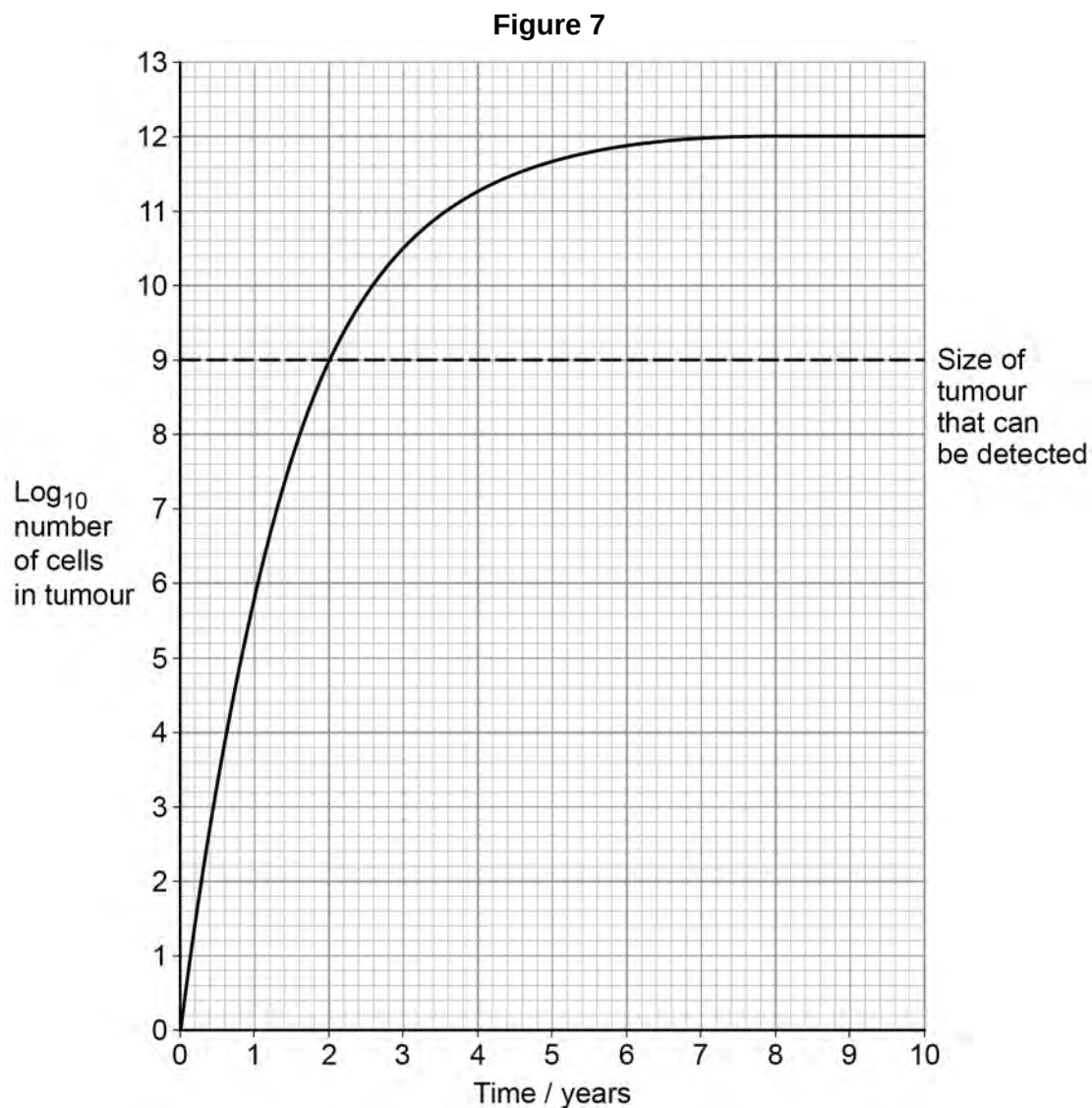


06

Uncontrolled cell division can cause tumours to form.

Figure 7 shows the growth pattern followed by a type of tumour.



06.1

Use **Figure 7** to calculate the percentage of maximum growth this type of tumour reaches before it can be detected.

You will need to use the 10^x button on your calculator.

[1 mark]

Answer = _____ %

0	6	.	2
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Figure 7 can also be used to calculate the age of this type of tumour.

At diagnosis, a patient had a tumour of 3.98×10^{11} cells.
Calculate the age of the tumour.

You will need to use the $\log_{10}$ button on your calculator.

[1 mark]

Answer = _____ years

Question 6 continues on the next page

Trexall is a drug that can be used to slow the development of various forms of cancer.

Trexall slows cell division by interacting with an enzyme called dihydrofolate reductase (DR).

DR is involved in making nucleotides; the substrate for DR is folic acid.

Figure 8 shows the chemical structure of Trexall.

Figure 9 shows the chemical structure of folic acid.

Figure 8

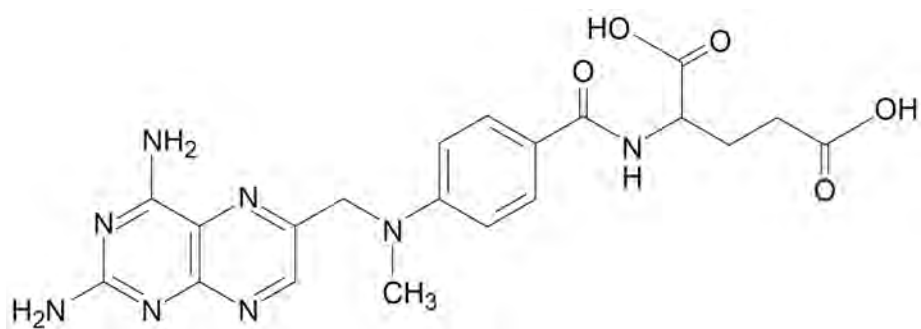
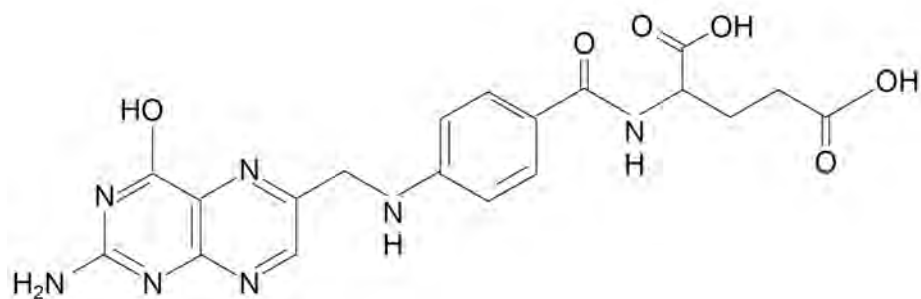


Figure 9



0 6 . 3

Use the information provided to suggest how Trexall slows cell division.

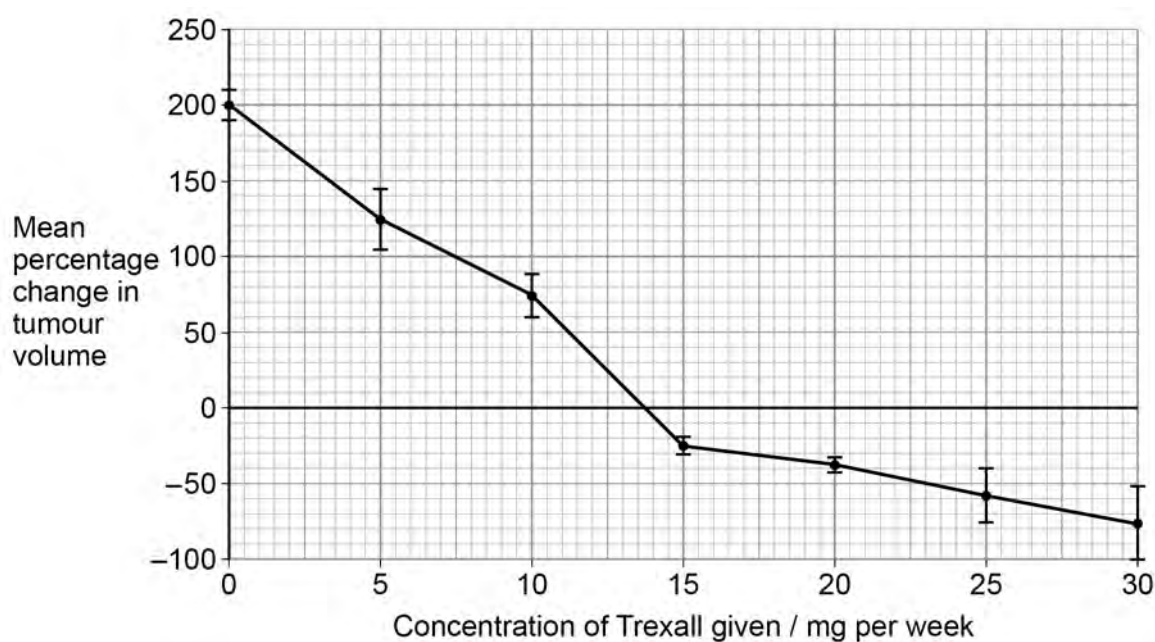
[3 marks]

Question 6 continues on the next page

Doctors investigated how the concentration of Trexall given to patients affected the growth of lung tumours. The doctors measured the volume of tumours at the beginning of the study and after 8 months.

Figure 10 shows the results of this investigation. The bars represent ± 2 standard deviations. A value of ± 2 standard deviations from the mean includes over 95% of the data.

Figure 10



0 6 . 4 The scientists measured the percentage change in tumour volume.

Suggest why they recorded both percentage change **and** tumour volume.

[2 marks]

Percentage change _____

Tumour volume _____

0 6 . 5

A lung cancer patient received 15 mg of Trexall per week. After treatment, the diameter of his lung tumour was 35.8 mm

Assuming the tumour was spherical, use the mean percentage change in tumour volume shown in **Figure 10** to calculate the volume of the patient's tumour **before treatment** with Trexall.

The formula for the volume of a sphere is $\frac{4}{3}\pi r^3$ where $\pi = 3.14$

[2 marks]

Answer = _____ mm³

0 6 . 6

To reduce the size of tumours, would it be better to use 30 mg of Trexall per week, or 20 mg of Trexall per week?

Explain your answer.

[2 marks]

Question 6 continues on the next page

Turn over ►

Trexall can also be used to slow the development of rheumatoid arthritis (a pain-causing joint disease).

Scientists investigated the effectiveness of Trexall as a pain relief treatment in 12 rheumatoid arthritis patients. All of the patients were female. They randomly divided the patients into two groups:

- Group **R** received Trexall tablets for 35 days
- Group **S** was a control group.

They asked both groups to rate their pain on a scale of 0–10 (0 being no pain and 10 being the worst pain possible) at the start and then every 7 days for 35 days. They calculated mean scores for each group.

Their results can be seen in **Table 3**.

Table 3

Number of days of treatment	Mean score for severity of pain (scale 0–10)	
	Group R	Group S
0	9.7	9.8
7	8.2	9.1
14	8.4	8.6
21	7.6	7.2
28	6.3	7.5
35	5.1	7.8

0 6 7

Apart from age and general health, give **two** important factors when choosing patients for this investigation.

[1 mark]

1 _____

2 _____

0	6	.	8
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A student analysed **Table 3** and concluded that Trexall was effective in reducing pain in arthritis patients.

Evaluate the student's conclusion.

[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

15

Turn over for Section B

Section B

Answer **one** question.

You are advised to spend no more than 45 minutes on this section.